

**LINUX
SCENARIO
BASED Q&A
DAY 8**

Q1. Deleted User Still Able to Login?

🧩 Scenario:

You've deleted a user account, but surprisingly, they can still SSH into the server.

🧠 Possible Root Cause:

The user's SSH key remains in the `~/.ssh/authorized_keys` file, and SSH does not strictly rely on `/etc/passwd` if the session is still valid or cached.

🔧 Solution:

1. Remove the user's home directory:
`sudo rm -rf /home/username`
2. Terminate any cached login sessions:
`sudo loginctl terminate-user username`

✅ Tip:

Always clean up the home directory and terminate sessions when deleting a user to prevent lingering access.

Q2. User Gets Disconnected Immediately After Login

🧩 Scenario:

The user successfully logs in via SSH or terminal but is instantly logged out or disconnected.

🧠 Possible Root Cause:

User startup files like `.bashrc`, `.bash_profile`, or `.profile` contain an `exit`, `logout`, or a command that triggers a fatal error.

🔧 Solution:

1. Review the user's startup scripts:
2. `~/.bashrc`, `~/.bash_profile`, and `~/.profile`
3. Look for unintended `exit`, `logout`, or any commands causing errors.

✅ Tip:

Always validate user shell scripts before applying them broadly, especially in shared or production environments.

Q3. Automatically Expire a User Account After a Set Date

🧩 Scenario:

You need to create a temporary user whose account should automatically expire after 7 days.

🧠 Possible Root Cause:

If account expiration is not set, temporary or contract users may retain access beyond their intended period, posing a security risk.

🔧 Solution:

1. Set an expiry date using the `chage` command:
`sudo chage -E $(date -d "+7 days" +%Y-%m-%d) username`

✅ Tip:

Use `chage -l username` to check the account's expiry details at any time.

Q4. Changes to systemd Service File Not Taking Effect

🧩 Scenario:

You've modified a .service file, but the service continues to behave as before.

🧠 Possible Root Cause:

Systemd caches service unit files in memory. After edits, it must be reloaded to apply changes.

🔧 Solution:

1. Reload systemd manager configuration:

```
sudo systemctl daemon-reload
```

2. Restart the service:

```
sudo systemctl restart myapp
```

✅ Tip:

Run `systemctl show <servicename>` to confirm that updated settings are in effect.

Q5. Service Enabled but Doesn't Start at Boot

🧩 Scenario:

You've run `systemctl enable servicename`, but the service is inactive after reboot.

🧠 Possible Root Cause:

- Service bound to the wrong target
- Missing or misordered dependencies
- Service starts before required resources are available
- Early failure without automatic retry

🔧 Solution:

1. Check boot target dependencies:

```
systemctl list-dependencies multi-user.target
```

2. Ensure proper startup order by adding `After=` and `Requires=` directives in the service unit.

3. Review boot-time service logs:

```
journalctl -b -u servicename
```

✅ Tip:

If your service depends on network or remote mounts, align it with `network.target` or `remote-fs.target` to ensure proper timing.

Q6. Enforce Password Reset on First Login

🧩 Scenario:

You want users to change their password the first time they log in.

🧠 Possible Root Cause:

By default, newly created users or manually set passwords remain active until explicitly changed. Users may continue using the default password without enforcing a reset, which is a security risk.

🔧 Solution:

1. Force immediate password change:

```
sudo chage -d 0 username
```

✅ Tip:

Recommended practice when setting up new user accounts in corporate environments.

Q7. systemctl restart Hangs Indefinitely

🧩 Scenario:

Restarting a service causes the command to hang without completing.

🧠 Possible Root Cause:

- The service's stop logic (ExecStop) is not implemented correctly.
- A stale or incorrect PIDFile prevents systemd from managing the process cleanly.

🔧 Solution:

1. Forcefully stop the service:

```
sudo systemctl kill servicename
```

2. Review the .service file for issues in ExecStop or PIDFile directives.

✅ Tip:

- Set a timeout to avoid hanging:
TimeoutSec=30
This ensures services don't stall the system during restarts.

Q8. Service Status Shows "inactive (dead)" After Start

🧩 Scenario:

You manually start a service, but systemctl status shows it as inactive (dead).

🧠 Possible Root Cause:

- The service process exits immediately after starting.
- Incorrect Type= directive in the systemd unit file.
- Misconfigured ExecStart or missing PID handling for forking services.

 Solution:

1. Set the correct Type in the unit file:
 - Use Type=simple for foreground processes.
 - Use Type=forking for background daemons that fork.
2. Ensure ExecStart points to the actual long-running process.

 Tip:

Forking services require a valid PIDFile and proper tracking to be managed correctly by systemd.

Q9. Daemon Running but systemctl status Shows “failed”

 Scenario:

The service is actively running, yet systemctl status reports it as failed.

 Possible Root Cause:

- The service was started manually (e.g., via nohup or a script), bypassing systemd.
- The PIDFile is missing, incorrect, or not maintained.
- Systemd is unable to track the process due to misconfiguration.

 Solution:

- Always manage services through systemctl to ensure proper tracking.
- If using a PIDFile, verify its accuracy and ensure the unit file points to it correctly.
- If supported, use Type=notify and implement sd_notify() in the daemon for reliable state reporting.

 Tip:

Avoid starting critical services manually; systemd relies on consistent tracking to manage state and dependencies effectively.

Q10. System Boot Slows Down After Enabling a Service

 Scenario:

System boot time noticeably increases after enabling a custom service.

 Possible Root Cause:

The service is blocking boot due to slow startup, often waiting on external dependencies like network or remote file systems.

 Solution:

1. Set a startup timeout in the unit file:
TimeoutStartSec=30
2. Ensure proper startup order by adding:
After=network-online.target or After=remote-fs.target

 Tip:

Use systemd-analyze blame to pinpoint which services are slowing down the boot process.